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Socratic ChatGPT: Theory, Design, and Empirical Evaluations

Short Paper

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Abstract

This study investigates the integration of Socratic teaching methods in generative AI chatbots and its impact on enhancing student learning experiences and outcomes. Utilizing the Design Science Research (DSR) model, we propose a Socratic ChatGPT that employs questioning techniques. The research aims to address the current limitations of AI chatbots in educational settings, such as the risk of students' over-reliance on direct answers, by fostering deep thinking and active learning. A field experiment, supplemented by surveys and interviews, is adopted to evaluate the effectiveness of these chatbots. Although results are pending, this study promises to offer valuable insights into the potential of AI chatbots to revolutionize learning processes and provide practical implications in educational settings.

Keywords: Generative AI Chatbots, Socratic Teaching Method, Design Science Research (DSR)

Introduction

Generative Artificial Intelligence (AI) chatbots, such as ChatGPT, are revolutionizing education, shaping the future of knowledge dissemination and acquisition, leading to lively debates (Diederich et al., 2022). Students can engage in interactive conversations with ChatGPT to seek explanations, clarification on challenging concepts, and additional insights on various subjects. By providing instant, personalized support anytime and anywhere, ChatGPT empowers students to take charge of their learning experience, promoting a proactive and inquisitive approach to education (Kasneci et al., 2023; Zhang et al., 2023). On the flip side of the coin, concerns tend to eclipse the commendations regarding the utilization of ChatGPT in educational settings, particularly in K-12 environments (Sok & Heng, 2023). In fact, some educational institutions outright prohibit the use of GPT, citing concerns such as the potential for plagiarism and the dissemination of misinformation, fostering a form of "learning without thinking" (Lo, 2023). Specifically, educators are apprehensive that over-reliance on ChatGPT could potentially hinder the cultivation of critical research and problem-solving skills. By delivering just an instant answer, ChatGPT might inadvertently discourage the development of essential educational goals of nurturing creativity and critical thinking skills such as exploring diverse resources, verifying information, and fostering independent thinking.

Despite the absence of a definitive resolution to this ongoing debate, the rapid integration of ChatGPT into educational settings appears inevitable, underscoring the urgent need for fine-tuning GPT to mitigate existing drawbacks. To harness the full potential of ChatGPT as an educational tool, it is imperative to refine its capabilities, encouraging a balance between instant responses and the promotion of essential cognitive skills. This proactive approach will not only mitigate the current drawbacks but also empower students to navigate information landscapes more effectively, leading to a deep understanding of the concepts and fostering a culture of independent inquiry and critical analysis.

Drawing from education literature of Socratic teaching method, we propose that the integration of Socratic learning, a dialogue-based educational method, into the design of ChatGPT might provide a promising avenue to promote critical thinking skills and active learning. The Socratic teaching method, characterized by inquiry-based learning through questioning and discussion, emphasizes a dynamic exchange of ideas and encourages active participation through dialogue (Dillon, 2016). By incorporating this approach, ChatGPT could be designed to not only provide information but also engage users in meaningful conversations, posing questions, guiding discussions, and prompting users to think critically about the subject matter (Dan et al., 2023). This dialogical learning experience has the potential to enhance user engagement, deepen comprehension, and foster the development of independent critical thinking skills, thereby enriching the educational value of ChatGPT. Following the Design Science Research (DSR) paradiagm (Abbasi & Chen, 2008), we initially identified a critical issue within current generative AI chatbots. This issue involves the tendency to provide direct answers to queries without prompting students to engage in critical thinking. Recognizing this as our target problem, we analyzed the meta-requirements to address this issue based on Socratic learning (i.e., curating Socratic-style prompts, encouraging interactive responses, and incorporating diverse topics). Subsequently, we identified the meta-designs to realize the above meta-requirement and developed a Socratic ChatGPT accordingly.

To empirically evaluate the effectiveness of our design, we employed a mixed-methods approach, incorporating historical learning records mining, a field experiment, surveys, and interviews to assess the impact of our Socratic ChatGPT. The two-month field experiment was conducted in collaboration with six classes in a middle school, employing a two-by-two design. The first experimental factor focused on whether students received call-to-action encouragements at the experiment's outset. Specifically, half of the students were randomly assigned to receive a call-to-action message along with a free account to encourage adoption of ChatGPT. This encouragement design served as an instrumental variable to address self-selection biases in adoption, particularly in exploring the effects of ChatGPT on learning outcomes. The second experimental factor involved the version of GPT used; students in the treated group utilized the Socratic ChatGPT, while those in the control group used the baseline version of ChatGPT. We collected data on their learning performances before and after the experiment to evaluate the performance changes. Meanwhile, we also conducted pre- and post-surveys to evaluate their perceptions of learning procedures, and supplemented our findings with insights from interviews to gain a deeper understanding of how Socratic ChatGPT influenced their learning processes.

We believe that this ongoing research contributes to the field in several ways. Firstly, it provides evidence-based insights into the advantages and disadvantages of generative AI chatbots in educational settings, thereby advancing ChatGPT research within educational contexts. Secondly, it introduces an innovative method to address the primary concerns associated with employing ChatGPT in educational settings. This contribution extends to the design of technological artifacts, as the study involves refining and evolving ChatGPT from its baseline version into Socratic versions. Meanwhile, for educational stakeholders, our research has the potential to improve the learning process, pushing the boundaries of generative AI chatbots in education. By improving learning performances and cultivating active learning, it contributes to ongoing efforts to enhance the educational experience.

Literature Review

Generative AI in education

The use of generative AI chatbots in education has been a topic of increasing interest in recent years (Okonkwo & Ade-Ibijola, 2021). Chatbots were initially designed as interactive computer programs that communicate using natural language or text, often powered by artificial intelligence algorithms to generate responses (Hussain et al., 2019). And their application and complexity have evolved since then. Previous

research highlights that Generative AI Chatbots, specifically, have the potential to dynamically generate tailored educational content, enabling real-time, personalized learning experiences and fostering a deeper understanding through adaptive, context-aware interactions (Kasneci et al., 2023; Zhang et al., 2023). However, their practical application has faced challenges and yielded mixed results.

The existing research indicates a cautious acceptance of chatbots in educational settings. For example, students are generally hesitant to fully embrace interactions with chatbots, often preferring human counterparts over automated agents (Durall Gazulla et al., 2023). This reluctance is further compounded by issues such as poor design and the inability of these chatbots to provide adequately personalized responses (Hwang & Chang, 2023; Smutny & Schreiberova, 2020). Furthermore, Fryer et al. (2017) highlighted instances where students showed reduced interest in language courses that were facilitated by chatbots(Fryer et al., 2017). This emphasizes the gap between the expected and actual effectiveness of these chatbots in educational settings. Despite these challenges, recent literature suggests a potential shift in the application of AI chatbots. Kohnke et al. (2023) examine the capabilities of ChatGPT for language instruction, acknowledging its advantages while presenting debates and drawbacks (Kohnke et al., 2023). Similarly, Ilieva et al. (2023) propose a theoretical framework for blended learning with chatbot integration, aiming to revitalize higher education pedagogy (Ilieva et al., 2023). In conclusion, the growing interest in generative AI chatbots in education represents a complex interplay between technological innovation and traditional teaching methods. Although current applications suggest a need for more nuanced and effective design strategies to improve student engagement and learning outcomes, the evolving nature of these AI systems presents exciting opportunities for the future.

The Socratic Teaching Method

The Socratic teaching method, a renowned pedagogical approach, is fundamentally rooted in the historical Socratic questioning depicted in Plato's ancient dialogues (Boghossian, 2012; Dan et al., 2023). The Socratic method prioritizes interactive learning by employing learner-centered questioning techniques. Instead of providing direct information and facts, instructors ask a series of open-ended, thought-provoking questions and encourage students to use their own words to define and discuss concepts (Edelson, 1996; Schneider, 2013). Specifically, Socratic teaching method mainly includes three key dimensions, namely focused questions, exploratory questions, and spontaneous questions (Paul & Elder, 2019). Focused questions are used by teachers to initiate and conclude dialectic interactions with students. These questions aim to direct students' attention to specific concepts or themes at the beginning of the conversation and guide them to summarize their thoughts and formulate final answers at the end. Exploratory questions are employed to inspire students to expand their thinking into different dimensions and explore a variety of ideas and possibilities during the middle of the conversation. These questions are designed to promote deeper thinking and extend the boundaries of students' learning. Spontaneous questions, on the other hand, are posed instantly during the dialogue based on students' responses. They are intended to keep the dialogue fluid and dynamic, encouraging students to remain actively engaged in the thinking process. Through these three types of questions, the Socratic method effectively maintains student attention, reinforces meaningful communication, and facilitates the internalization of learning (Dillon, 2016).

Existing research also confirms that the Socratic method has been effective in improving learning attitudes, higher-order thinking skills, self-reflection, and overall performance (Dinkins & Cangelosi, 2019; Lee, 2014). However, most of the literature assumes that dialogue is conducted through speech in face-to-face interactions. The effect of combining this method with the design of generative AI chatbots remains unexplored (Le, 2019). Traditionally, the use of Socratic questioning techniques has been directly managed by teachers, enabling them to readily identify students' problems. Consequently, teachers can pose the appropriate questions, facilitating a smooth transition into deep learning. However, in interactions between generative AI chatbots and students, the initiative to ask questions typically rests with the students. If a student is unsure of what to ask or fails to communicate their situation, it becomes challenging for ChatGPT to pose the right questions. This can result in ineffective dialogues, potentially leading to student frustration and disengagement. Current studies have not yet addressed these nuances. Therefore, as we transition from interpersonal to human-computer communication, it is essential to further explore which aspects of the Socratic method are effective in ChatGPT applications and how to refine the questioning techniques to optimize their effectiveness.

System Design

To create a refined Socratic version generative AI chatbots, we adopt the design science research framework (Abbasi & Chen, 2008). We first selected the Socratic teaching method (STM) as the kernel theory. Followed the theory of STM, we next defined the meta-requirements of the dialogue style design for the evolved generative AI chatbot. This allows the chatbot to use questioning techniques to assist students in refining their ideas and encourages active thinking. To fulfill those needs, we also proposed meta-designs. Finally, we instantiated a Socratic ChatGPT that carries out the meta-designs. Our design framework is outlined in Table 1.

Table 1. A Design Framework for Socratic ChatGPT

Research goal	Improve the design of generative AI chatbots to facilitate student learning and enhance human intelligence.
Kernel theory	Based on the Socratic method, the extensive use of questions could stimulate students' cognitive engagement and foster their active thinking, and thus enhance the learning performance. Therefore, in this study, we can train Chatbot to apply a variety of questioning techniques to avoid the harm that chatbot directly provides students with answers to their thinking ability.
Meta-requirements	1. Tailor interactions by understanding each student's background and learning preferences, posing focused questions to encourage students thinking. 2. Use exploratory questions to prompt students to think about their questions from diverse perspectives, aiding in deeper comprehension. 3. Encourage students to use their own words to integrate different viewpoints, fostering the internalization of information, and aiding them in developing their own well-considered conclusions.
Meta-design	1. Socratic ChatGPT integrates historical learning records mining, featuring a dynamic prompt system that adapts to individual student's learning styles and paces. 2. The chatbot's design focuses on Socratic questioning techniques, breaking them down into specific dialogue tactics for effective interaction and critical thinking encouragement. 3. Enhanced training prompts guide students to synthesize information from various sources, encouraging them to think independently and express their own opinions through Socratic questioning.
Testable hypotheses	H1: Compared to not using a generative AI chatbot, using a chatbot results in a better learning performance and experience for students. H2: Compared to using a baseline ChatGPT, using a Socratic ChatGPT results in a better learning performance and experience for students.
System instantiation	Instantiation of the meta-design artifacts and implementation of the designed system.
Evaluation	Empirical evaluation of the testable hypotheses via a mixed-methods approach, incorporating historical learning records mining, a field experiment, surveys, and interviews.

Meta-requirements

Based on the principles of the Socratic teaching method and the requirements for enhancing educational chatbots, here are three detailed meta-requirements for the Socratic ChatGPT:

1. Identify and understand students' background information and needs, make the conversation more customized and personalized, thereby encouraging students to ask key questions. Specially, a key principle of the Socratic teaching method is student-centered questioning. Therefore, the first requirement is to

understand the student's background information, such as previous academic performance, learning style, learning preferences, etc. It should be capable of tailoring conversations based on these insights, encouraging students to raise critical questions relevant to their unique learning journey. The system must adaptively modify its interaction style, aligning with the student's educational level, interests, and current understanding, thus fostering a more personalized and engaging learning experience.

2. Use Socrates' questioning techniques to help students refine and improve their ideas, encourage them to ask and think about their own problems from different perspectives. According to the Socratic method of teaching, instructors should ask a series of open-ended, thought-provoking questions instead of providing direct information and facts. Therefore, in order to help students think deeply, Socratic ChatGPT should be proficient in various questioning techniques, such as exploratory questions and spontaneous questions to promote clarification of ideas, and guide students to think about problems from multiple dimensions.

3. Use questioning and guidance methods to encourage students to integrate answers from different perspectives and internalize them into their own understanding. Socratic teaching method encourages students to reformulate their understanding through multiple rounds of dialogue, thereby deepening their comprehension of the specific topic. Therefore, Socratic ChatGPT should foster an environment where students can synthesize different viewpoints and information sources, guiding them to form their own informed conclusions. The chatbot's responses and prompts should be geared towards refining students' critical thinking abilities, enabling them to analyze, compare, and contrast varying opinions and data points.

Meta-design

To meet the meta-requirements, we obtained access to OpenAI's API services. This enabled us to deploy a tailored ChatGPT model within our computational framework for targeted experimentation (OpenAI, 2024). We then implemented prompt engineering techniques to weave the Socratic teaching method into the ChatGPT. Specifically, we designed prompts rooted in the principles of Socratic pedagogy to steer the model toward generating responses that demonstrate both depth of inquiry and breadth of thought. Here are three corresponding meta-design methods:

1. Incorporating with historical learning records mining, Socratic ChatGPT was designed with a dynamic prompt system that not only retrieves and integrates students' historical learning records and preferences but also adapts to their unique communication styles and learning paces. This system analyzes students' previous interactions and academic weaknesses, using this data to tailor prompts that encourage students to articulate and explore their questions more deeply.

2. The chatbot's interaction design was centered on Socratic questioning methods. We deconstructed the Socratic questioning into specific conversational tactics to ensure effective incorporation within the chatbot dialogues. Those techniques involve prompting students to articulate and clarify their thoughts, challenging their presumptions, and guiding them towards exploring the profound implications and potential solutions of their inquiries. By providing extensive information on Socratic questioning techniques, including different types and examples of questions to ask, we trained Socratic ChatGPT in guiding students to ask questions and think from different perspectives.

3. To encourage students to an integrated understanding, we have enhanced the design of the training prompt. This prompt highlighted a specialized feature, which guide students through a process of synthesizing information gathered from various sources and viewpoints. After recording multiple rounds of conversations, ChatGPT should use Socratic questioning to guide students to think and express their own opinions.

Testable Hypotheses

Previous studies have demonstrated that generative AI chatbots are capable of providing customized responses to specific queries from students. In addition, they can also provide support and assistance to students regardless of time and space constraints. Therefore, we propose the following hypotheses for the effect of learning with *a generative AI chatbot:*

H1: Compared to not using a generative AI chatbot, using a chatbot results in a better learning performance and experience for students.

As discussed previously, the Socratic chatbot, based on the Socratic method, fosters critical thinking and self-directed learning, improving comprehension and engagement. Its personalized interaction caters to individual learning styles. Conversely, a baseline version AI chatbot may lack depth and adaptability. Hence, a Socratic chatbot could offer superior learning outcomes and a more engaging educational experience. Thus, we propose that:

H2: Students using a Socratic ChatGPT will have a better learning performance and positive learning experience compared to those who use a baseline ChatGPT.

System Instantiation and Study Context

As discussed above, we designed baseline ChatGPT and Socratic ChatGPT for instantiation. As shown in Figure 1, Socratic ChatGPT firstly gives the response to clarifying student needs. Meanwhile, it proactively asks questions based on students' needs, guiding them to think more deeply, thus better support them in solving their own problems. After several rounds of questions and answers, students will be guided to summarize their own understanding and answers. While the baseline ChatGPT provides answers directly. Then, to evaluate the effectiveness of the abovementioned design, we collaborated with one middle school located in Hefei, Anhui Province, China. The study involved providing educational information about the use of generative AI chatbots for students. We provided a unique link for each student to access the designed generative AI Chatbot. During the whole winter holidays, students are allowed to use generative AI Chatbots to do any learning tasks such as helping them write essays, revising essays, looking up English facts, answering math questions and so on.

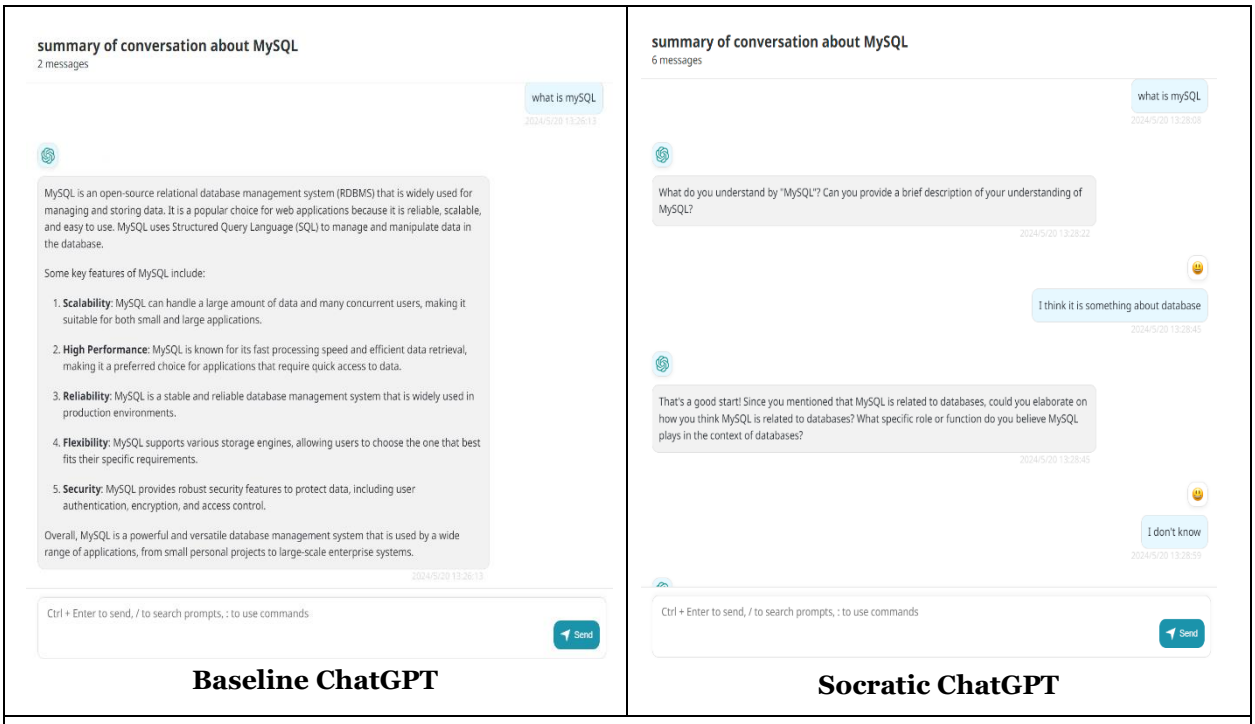


Figure 1. Demos of Baseline ChatGPT and Socratic ChatGPT

Empirical Evaluation

Research Design and Procedure

We conducted a 2 (with usage remind vs. without usage remind) × 2 (Baseline ChatGPT vs. Socratic ChatGPT) factorial field experiment during the winter holidays from January 26 to March 1, 2024. Our participants comprise students from six different classes of year 8 and 9.

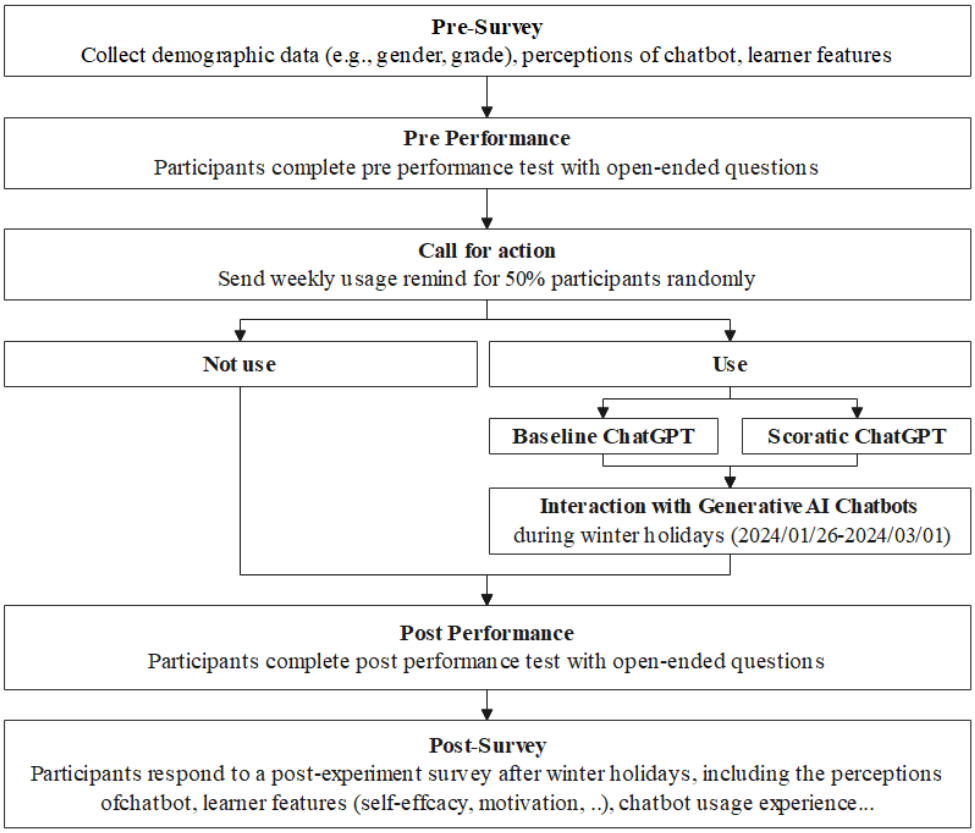


Figure 2. Flow chart of Experimental Procedures

The experimental procedure is outlined in Figure 2. Prior to the experiment, we conducted a pre-survey, where we gathered demographic information of the learners (see Table 2), such as their gender, years, and final exam scores. We also measured their learning attitudes, such as learning motivation, performance orientation, self-efficacy, and perceptions of generative AI chatbots (Santhanam et al., 2008, 2016; Tseng et al., 2023). Subsequent to the initial phase, we administered an online pre-test comprising various exploratory open-ended questions. This step was pivotal in establishing the learners' performance baseline prior to the experimental intervention. Following this, students were randomly assigned to receive weekly usage reminders or not. Specifically, half of the students were randomly chosen to receive a call-to-action message to stimulate their usage of Chatbot. This approach aims to mitigate self-selection bias and determine whether the usage of a generative AI chatbot affects students' academic performance. Then, to ensure consistency in teaching content, style, and other variables, we performed randomization at the class level to randomly assign each class to either the baseline or Socratic version ChatGPT. Students in the Socratic group utilized the Socratic ChatGPT, while those in the baseline group used the baseline version of ChatGPT. The experimentation took place during the onset of winter holidays, and each student was provided a unique access code for their respective chatbot version. This setup was designed to reduce the likelihood of information exchange among students. Students were encouraged to utilize these chatbots for various tasks, including delving into the answers to pretest questions. After the experiment, we conducted a post-test using the same exploratory open-ended questions to evaluate the learners' performance changes. A comprehensive post-survey was also conducted to gauge the students' experiences and the educational shifts engendered by the chatbot usage. For the analysis of the collected data, a two-way ANOVA approach was planned. Moreover, selected participants will be invited for structured interviews, aimed at providing a deeper interpretation of the quantitative findings.

Table 2. User Characteristics in the Background Survey			
	Category	Freq.	Percent
Final exam scores	A	55	19.8
	B	86	30.9
	C	88	31.7
	D	33	11.9
	E	16	5.8
Year	Year 8	97	34.9
	Year 9	181	65.1
Gender	Male	136	48.9
	Female	142	51.1

Concluding Remarks

This research examines the impact of generative AI chatbots on traditional education and their influence on the student learning process. Additionally, it explores the design of an optimal generative AI chatbot that incorporates Socratic teaching methods to achieve the better educational outcomes. The study found that traditional AI chatbots, which typically provide direct responses, may limit students' depth of thought and cognitive engagement. In response, we employed the Socratic method, which is characterized by inquiry-based learning through questioning and discussion, to develop an innovative chatbot variant. This approach not only fosters student autonomy but also stimulates critical thinking skills.

Through a mixed-method study that combined experimental design and interviews, we evaluated the impact of a new chatbot design on students' learning experiences and outcomes. Our study provides valuable empirical data for the field of educational technology. Theoretically, it advances the understanding of AI's transformative potential in education, challenging existing paradigms and suggesting new directions for AI integration in learning environments. Our research provides direction for optimizing AI design in educational tools, guiding future AI design in educational tools to enhance learning processes and outcomes. It not only deepens the theoretical understanding of AI's role in education but also makes a significant practical contribution.

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